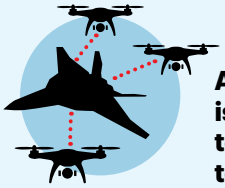
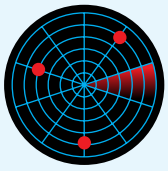
HOW THE NEW GENERATION WORKS

Key features of sixth-gen fighter jets



SYSTEM-OF-SYSTEMS

A sixth-generation stealth jet is a command node in an interconnected combat ecosystem that can unite and steer low-cost drone swarms and advanced air defence systems



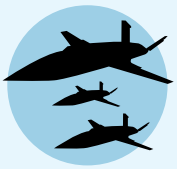
ENHANCED STEALTH

It will combine advanced low-observable shaping, radar-absorbent materials, infrared signature suppression and fully internal weapons carriage, minimising radar and thermal detection



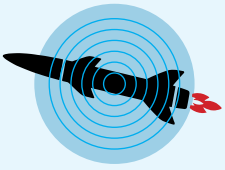
SENSOR FUSION WITH AI

Advanced AESA radars, EO/IR systems, passive sensors and EW suites link up to provide 360-degree battlespace picture. AI to support predictive threat analysis, autonomous decision-making, rapid target prioritisation



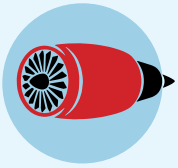
WINGMAN DRONES

In this key concept, the jet will function as an airborne command node, controlling swarms of loyal 'wingman' drones to be used for strike, decoy, reconnaissance



ADVANCED WEAPONRY

Next-gen beyond visual range missiles and hypersonic (exceeding Mach 5 speeds) strike systems. Ability to carry nearly double the weapons payload of America's fifth generation F-35A fighter



ADAPTIVE CYCLE ENGINES

These can operate under subsonic, transonic and supersonic conditions, and capable of producing over 40,000 pounds of thrust each. Expected to enable Mach 2+ supercruise, improved fuel efficiency

ROARS FROM THE PAST

Evolution of the fighter jet

↘ FIRST-GENERATION
Mid-1940s-mid-'50s: Subsonic speeds, no radar, armed with cannons, unguided bombs/rockets

AIRCRAFT: Me 262 (Germany), F-86 Sabre (US), Mig 15 (USSR)

↘ SECOND-GEN
Mid-'50s-mid-'60s: supersonic flight, afterburners, basic radar, air-to-air guided missiles

AIRCRAFT: F-100 Super Sabre (US), Mig-21 (USSR), Hawker Hunter (UK), Mirage III (France)

↘ THIRD-GEN
Early '60s-1970: Early beyond visual range missiles, avionics

AIRCRAFT: F-4 (US), Mig-23 (USSR)

↘ FOURTH-GEN
1970-late '80s: Head-up displays, fly-by-wire ability

AIRCRAFT: Mig-29 (USSR), F-16 (US), Mirage 2000 (France)

↘ 4.5-GEN
Late '80s-late '90s: Partial stealth, thrust-vector engines, AESA radar

AIRCRAFT: Sukhoi Su-35 (Russia), Rafale (France)

↘ FIFTH-GEN
2005-present: Full stealth, EW suites, sensors across airframe, fully networked capabilities, internal weapons bay

AIRCRAFT: F-22 Raptor, F-35 (US), Su-57 (Russia), J-20, J-35 (China)

assured market. "We need to be fully involved in all niche technologies and absorb them to our benefit. This will be our challenge and we must be the sole supplier on some hardware and software," says Air Marshal Nambiar.

Some observers believe such exposure could also support India's own long-term military aviation goals beyond the fifth-generation Advanced Medium Combat Aircraft (AMCA) project by speeding up technology development. Currently in the prototype stage, the AMCA could enter service by 2035 if timelines hold. In contrast, China already has two stealth fighters—J-20 and J-35—operationally deployed, with Pakistan reportedly set to acquire J-35s soon. Given the fact that fighter jet technology is undergoing upgrades every decade or so, India must be abreast of latest advancements, feel experts.

The GCAP and the FCAS aim to